Exercise 2

Website, Email, Whois, DNS, and Network Footprinting on the Target Network

Aim

To gather detailed technical information about the target's website, email servers, domain registration (Whois), DNS records, and network range using footprinting tools.

Theory

This exercise goes beyond Exercise 1 by examining technical infrastructure instead of public/social information: Website Footprinting, Email Footprinting, Whois Footprinting, DNS Footprinting (A/MX/NS/TXT/CNAME records), and Network Footprinting (IP range and route to the target).

Tools Required

whois, nslookup, dig, traceroute, WhatWeb/Wappalyzer, browser DevTools.

Target

scanme.nmap.org - the Nmap Project's own test host, publicly documented as open for scanning/recon practice, so the commands below were run live against a target that explicitly permits it.

Step 1 - Whois

```
$ whois scanme.nmap.org
% IANA WHOIS server
% for more information on IANA, visit http://www.iana.org
% This query returned 1 object

refer:        whois.publicinterestregistry.org

domain:       ORG

organisation: Public Interest Registry (PIR)
address:      11911 Freedom Drive,
address:      10th Floor, Suite 1000
address:      Reston VA 20190
address:      United States of America (the)

contact:      administrative
name:         Director of Operations, Compliance and Customer Support
organisation: Public Interest Registry (PIR)
address:      11911 Freedom Drive,
address:      10th Floor, Suite 1000
address:      Reston VA 20190
address:      United States of America (the)
phone:        +1 703 889 5778
```

```
fax-no:       +1 703 889 5779
e-mail:       ops@pir.org

contact:      technical
name:         Senior Director, DNS Infrastructure Group
organisation: Identity Digital Limited
address:      c/o Identity Digital Inc.
address:      10500 NE 8th Street, Suite 750
address:      Bellevue WA 98004
address:      United States of America (the)
phone:        +1.425.298.2200
fax-no:       +1.425.671.0020
e-mail:       tldtech@identity.digital

nserver:      A0.ORG.AFILIAS-NST.INFO 199.19.56.1 2001:500:e:0:0:0:0:1
nserver:      A2.ORG.AFILIAS-NST.INFO 199.249.112.1 2001:500:40:0:0:0:0:1
nserver:      B0.ORG.AFILIAS-NST.ORG 199.19.54.1 2001:500:c:0:0:0:0:1
nserver:      B2.ORG.AFILIAS-NST.ORG 199.249.120.1 2001:500:48:0:0:0:0:1
nserver:      C0.ORG.AFILIAS-NST.INFO 199.19.53.1 2001:500:b:0:0:0:0:1
nserver:      D0.ORG.AFILIAS-NST.ORG 199.19.57.1 2001:500:f:0:0:0:0:1
ds-rdata:     26974 8 2 4fede294c53f438a15
```

Observation: the .org TLD registry (PIR) referral record was returned, confirming registrar delegation (Public Interest Registry) and nameserver pool.

Step 2 - DNS Footprinting (nslookup, dig)

```
$ nslookup scanme.nmap.org
Server:             192.168.1.1
Address:     192.168.1.1#53

Non-authoritative answer:
Name:  scanme.nmap.org
Address: 45.33.32.156



$ dig scanme.nmap.org NS

; <<>> DiG 9.10.6 <<>> scanme.nmap.org NS
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 15748
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;scanme.nmap.org.          IN     NS

;; AUTHORITY SECTION:
nmap.org.           600    IN     SOA    ns1.linode.com. hostmaster.nmap.com.
2021000018 14400 14400 1209600 3600

;; Query time: 209 msec
```

```
;; SERVER: 192.168.1.1#53(192.168.1.1)
;; WHEN: Sat Oct 03 21:00:43 IST 2026
;; MSG SIZE  rcvd: 110
```

```
$ dig scanme.nmap.org TXT

; <<>> DiG 9.10.6 <<>> scanme.nmap.org TXT
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 26986
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;scanme.nmap.org.          IN     TXT

;; AUTHORITY SECTION:
nmap.org.           600    IN     SOA    ns1.linode.com. hostmaster.nmap.com.
2021000018 14400 14400 1209600 3600

;; Query time: 241 msec
;; SERVER: 192.168.1.1#53(192.168.1.1)
;; WHEN: Sat Oct 03 21:00:43 IST 2026
;; MSG SIZE  rcvd: 110
```

Observation: scanme.nmap.org resolves to 45.33.32.156 (Linode). The NS/TXT queries delegate to nmap.org's SOA (ns1.linode.com) - normal for a scoped subdomain test host.

Step 3 - Website Footprinting

```
$ curl -I http://scanme.nmap.org
HTTP/1.1 200 OK
Date: Sat, 03 Oct 2026 15:30:52 GMT
Server: Apache/2.4.7 (Ubuntu)
Accept-Ranges: bytes
Vary: Accept-Encoding
Content-Type: text/html
```

Observation: HTTP headers show Server: Apache/2.4.7 (Ubuntu) - an old version worth checking against known CVEs in a real engagement. (WhatWeb was not installed in this environment; 'whatweb scanme.nmap.org' is the documented Kali command for fuller fingerprinting.)

Step 4 - Email Footprinting

Not applicable to scanme.nmap.org (no contact/mail form exposed). Documented procedure: submit a test message via the target's contact form, then inspect the received email's full headers for the sending mail server IP and MTA software.

Step 5 - Network Footprinting (traceroute)

```
$ traceroute scanme.nmap.org
traceroute to scanme.nmap.org (45.33.32.156), 15 hops max, 40 byte packets
 1  192.168.1.1 (192.168.1.1)  3.271 ms  4.172 ms  1.988 ms
 2  192.168.0.1 (192.168.0.1)  5.162 ms  3.578 ms  2.962 ms
 3  * 10.208.0.1 (10.208.0.1)  6.963 ms
```

Observation: capture shows the first hops from the local network before the trace was stopped; a full run reveals every router hop to the target.

Expected Output

A consolidated document containing Whois registration details, DNS records, web server technology stack, and the network path to the target - all captured above from a live, scan-permitted host.

Viva Questions

1. What information does a Whois lookup provide?
2. What is the difference between an A record and an MX record in DNS?
3. What is a DNS zone transfer, and why is it a security risk if misconfigured?
4. How can viewing a website's page source help in footprinting?
5. What does traceroute show, and how is it useful to an attacker?